

Referee guide

Generic Identifiability of Strongly Tree-Child Level-2 Jukes–Cantor Networks

Claim and scope. The main theorem concerns all binary standard semi-directed S_{TC} level-2 networks under open four-state Jukes–Cantor parameters, with generic topology identification modulo ordinary triangle redirection T . Theorem 3.1 proves that no additional triangle-count hypothesis is needed. It classifies one-sided full-dimensional stochastic containment, not equality of complete stochastic images. The sharpness theorem places the explicit Theta pair in $W_{TC} \setminus S_{TC}$ and extends it to every $n \geq 4$. The reconstruction theorem returns a canonical structural topology modulo T ; it does not claim that a fixed input distribution lies in every redirected model.

Suggested proof-reading order.

1. Definitions and scope: Section 2 and Theorems 3.1–3.5.
2. Bridge localization: Proposition 4.1, Lemmas 4.2–4.3, and Theorem 4.4. The key issue is that reciprocal bridge gauges are complete and a regular source product cell projects to genuine local one-sided containments without assuming target regularity.
3. Cut preservation: Theorem 5.1. Check especially Lemma 5.4, where four wrong-split minors exclude a bridge joining two active endpoint tensors.
4. Local reconstruction: Proposition 6.1 and Theorem 6.6. These give the cycle/four-theta core list, rigid supports, one-port segment location, and two-port order recovery.
5. Finite atlas: Section 7. The load-bearing residual step is Theorem 7.5: eight role orbits, 192 labelled records, 1,686 valid completions, and strict trinet separators. The polynomial F_{abc} is attributed to Englander et al.; only its restored-support application is new.
6. Root and global synthesis: Theorem 8.2, Proposition 8.3, Corollary 8.4, and Proposition 8.6. The retained-arrowhead rerooting case and contextual T gluing deserve special attention.
7. Genericity and reconstruction: Lemma 9.1, Proposition 9.2, Theorem 9.3, and Remark 9.4. These connect full semialgebraic source dimension to a source-relative regular open set, pull the proper observed exceptional set back to a proper parameter polynomial, and distinguish structural reconstruction modulo T from input-specific triangle-orientation membership.
8. Sharpness: Section 10 and Appendix B. The exact quadratic point, all Fourier coordinates, two rank-eight Jacobians, leaf-substitution inverse, and $2n$ dimension defect are explicit.

Quick verification.

```
bash reproducibility/verify_quick.sh
```

This checks the archive manifest, the automatic-triangle certificate and its independent census, the underlying exact statistical release, publication scope/count audit, bibliography, PDF build, embedded fonts, and unresolved-reference/overfull-box conditions.

Full independent verification.

```
bash reproducibility/verify_full.sh
```

The full command independently regenerates the complete multi-triangle rooting universe in Python and C++, and verifies the unchanged cycle/theta algebra, directed relations, seven-port closure, and pointwise-cut replay against the preserved clean full-adversarial base transcript. The optional `verify_regenerate_all.sh` reruns those large algebra generators from scratch. The expected terminal line is

AUTHOR-READY FULL INDEPENDENT VERIFICATION PASSED

Theorem-to-artifact map.

Result	Principal commands/files
Automatic triangle bound	<code>publication/src/verify_multitriangle_exclusion.py</code> ; independent C++ rooting census
Five/six-port theta atlas	<code>publication/src/regenerate_nonroot_algebra.py</code> ; complete signature binaries and pair TSVs; independent adaptive-list reviewer
Cycle/cross-generator atlas	<code>publication/src/regenerate_cycle_algebra.py</code> ; relation summaries and complete TSVs
Seven-port closure	<code>exact_release/src/verify_seven_port_closure.py</code> ; independent <code>review_seven_port_closure.py</code>
Pointwise cuts	<code>exact_release/review/review_pointwise_cut.py</code> ; 177 endpoints, 453 one-active crossings, 547 Bernstein expansions
Root reduction	primary and independent scripts in <code>exact_release/src</code> and <code>exact_release/review</code>
Global synthesis	<code>build_sharp_boundary_certificate.py</code> ; <code>review_final_synthesis.py</code>
Theta sharpness	<code>verify_theta_sharpness.py</code> ; exact quadratic certificate

Adversarial questions.

- Does standard S_{TC} mean every admissible rooting, rather than the supplied rooted presentation? Yes; Section 2 fixes R_{TC} , W_{TC} , and S_{TC} separately.
- Is the former one-triangle condition still an assumption? No. Theorem 3.1 proves that every binary level-2 topology in W_{TC} automatically has at most one triangle per blob.
- Is containment oriented correctly? The finite relation stores source nonvanishing masks as subsets of target masks. Strict separators vanish on the source and have fixed nonzero target sign, which proves open-image disjointness.
- Could distant blobs compensate? The peeling chart makes the source germ a product modulo exactly the bridge gauges; projection gives each local containment separately.
- Is the seven-port count assumed? No. Both primary and independent programs regenerate the role orbits and every completed graph from primitive cores.
- Is a boundary specialization used as an interior witness? Boundary choices appear only to prove a polynomial nonzero. Every overlap or strict sign used in the stochastic theorem is certified in the open domain.
- Does T mean equal complete images, or that one input point belongs to every redirected model? No. The proof supplies a common full-dimensional regular germ and the algorithm returns the structural T -class; pointwise membership requires separate semialgebraic tests.

Ardiyansyah (2021) is included as direct historical context for Fourier-invariant distinguishability of simple and semisimple level-2 varieties.

The detailed crosswalk is `docs/THEOREM_CERTIFICATE_CROSSWALK.md`. Hashes verify byte integrity, not mathematics.